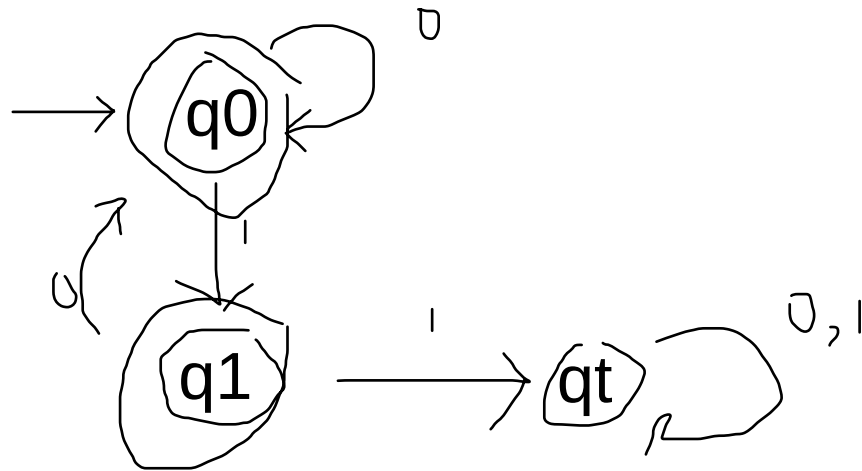


Theory of Computation

Exercise 2: (Deterministic Finite Automata - DFA)

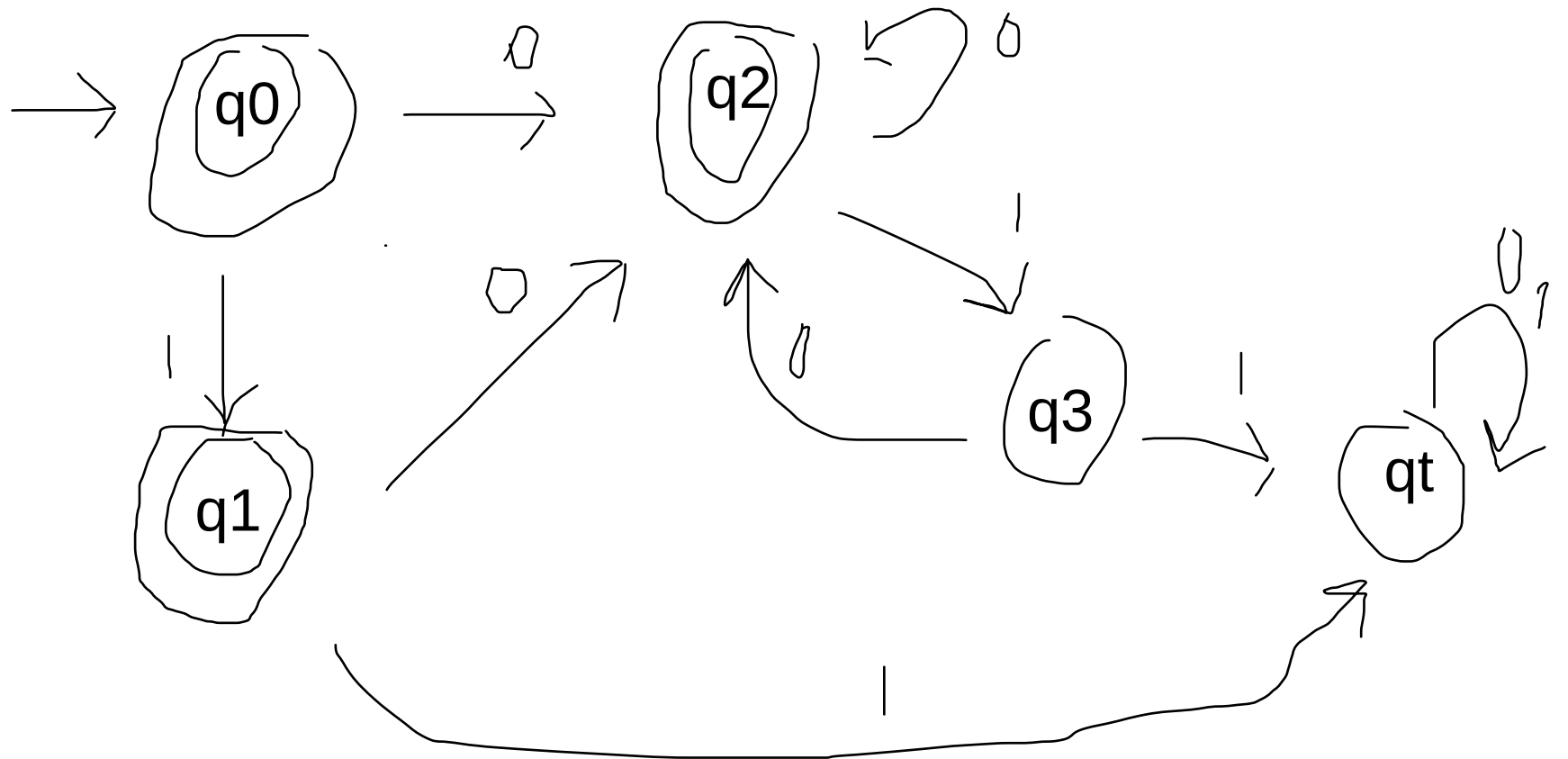
1. Draw DFA for $L1$

$L1 = \{w \in \{0,1\}^* : w \text{ has no substring } 11\}$



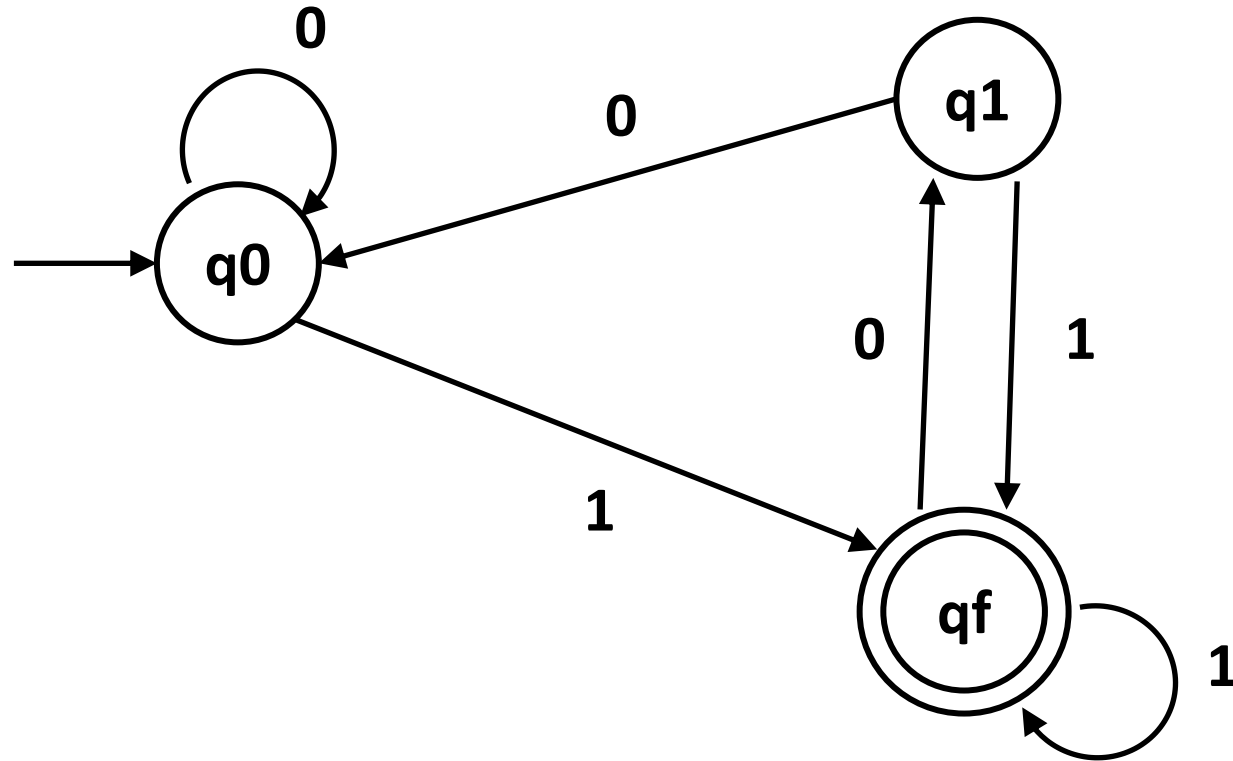
2. Draw DFA for L2

$$L2 = \{0, 10\}^* \cup \{1\}$$



3. Find the language of DFA M.

M:



$$L = \{ w \text{ in } \{0, 1\}^+ : w \text{ end with } 1 \}$$

* 4. Draw DFA for L3

(Homework 1)

$$L3 = \{ a^m b^n : m + n = 5; m \text{ and } n \geq 0 \}$$

